Advanced Statistics

19-01-26

data - weights of students

19.2, 30.4, 50.6, 70, 55, 42, 100.8 . . . . . .

- continuous data

Analysis - $\underbrace{\text{mean, median, mode}}_{\text{measures of central tendency}}$, $\underbrace{\text{std dev, variance, iqr}}_{\text{measures of dispersion}}$

mean = 48.7

Data Distributions

- last month expenses - 10,000 ✓
- Bank statement - 100, 2500, 14, 10, 5, 1200 . . . . .
  or
  payment history

Raw data

↳ you want to get more information from this raw data

→ you'll apply statistical methods to convert raw data into insights/useful information

mean value = 250 | 91 = 700

max = 4100 | 92 = 3000

min = 5 | std dev = 560

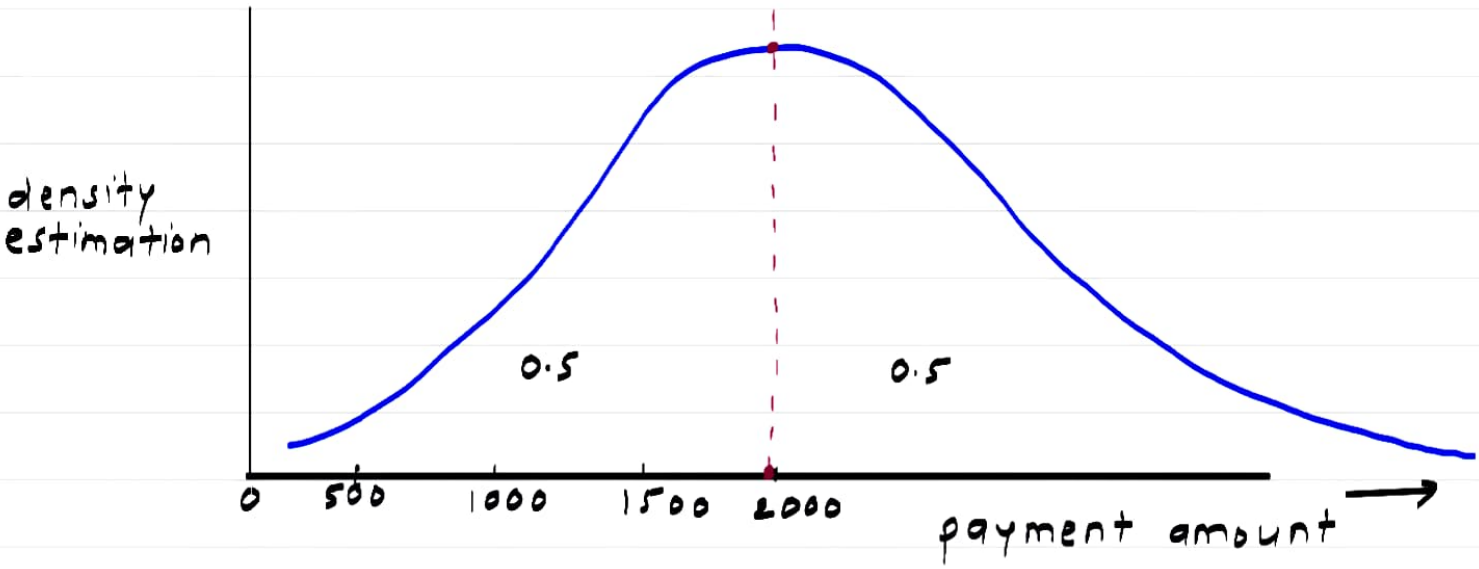
payments - continuous data

↓

visualize data using

1) histogram 2) Box plot 3) violin plot

4) KDE plot



no. of payments betn 500-1000 are less than

no. of payments betn 1500-2000

A data distribution represents how values of a variable in data are spread, how often they occur in given range, revealing patterns of variation

payment types - 15 - food 1500 - investment

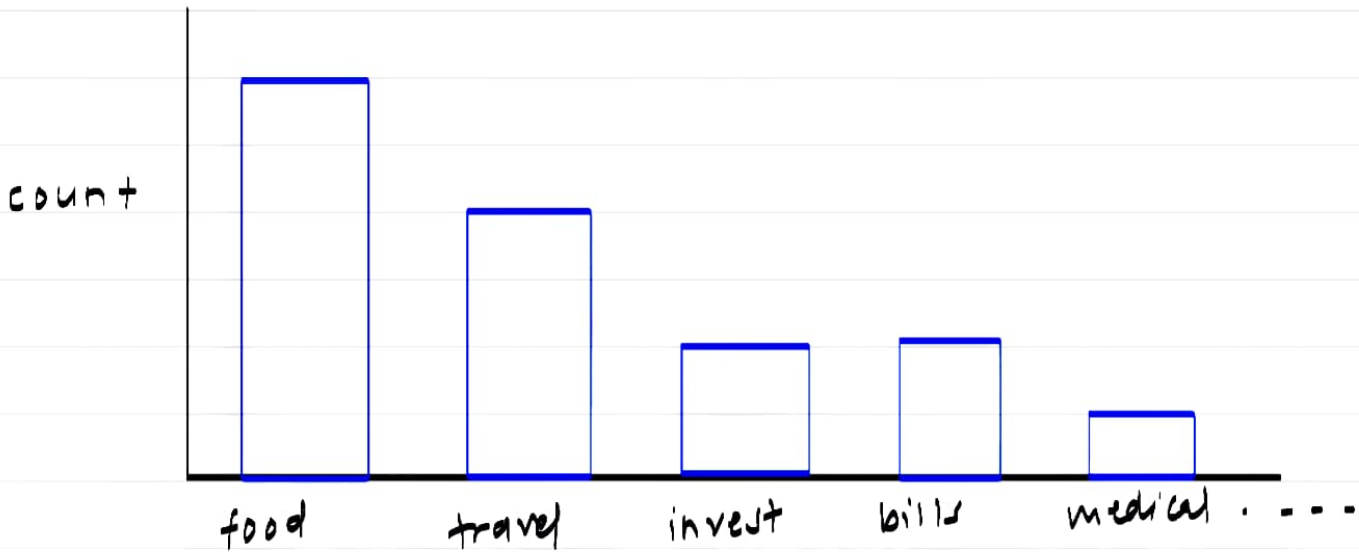
↓ 1000 - travel 2000 - rent . . . . .

categorical variable

- we can visualize no. of types of payments

total no. of payments = 150

food = 40
travel = 20
investment - 10
bills - 10
medical - 5
⋮



A data distribution is very powerful representation derived from data which provides huge amount of information about data

Using data distribution we can gain insights on
- shape of data distribution
- central tendencies
- dispersion
- outliers, extreme values
- patterns
etc.

Uses of data distributions

- Insight generation - for data driven decision making

- Data Interpretation : converting raw data into easy to understand visualization

- Predictive Analytics : computing probabilities of specific outcomes

Cricket match results for Team A vs Team B

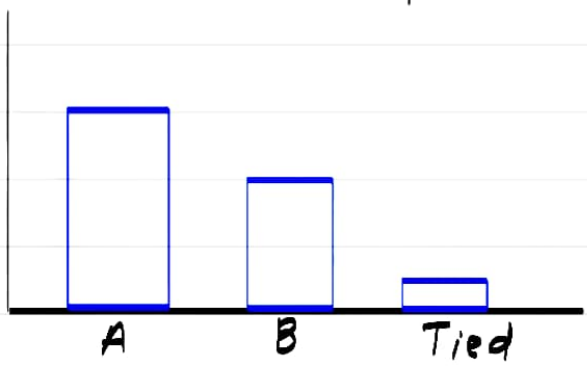
80 matches → A, B, A, A, B, B, B, T, A, T, B.....

Tommorrow there is a match betn A and B.
What's probability that A will win?

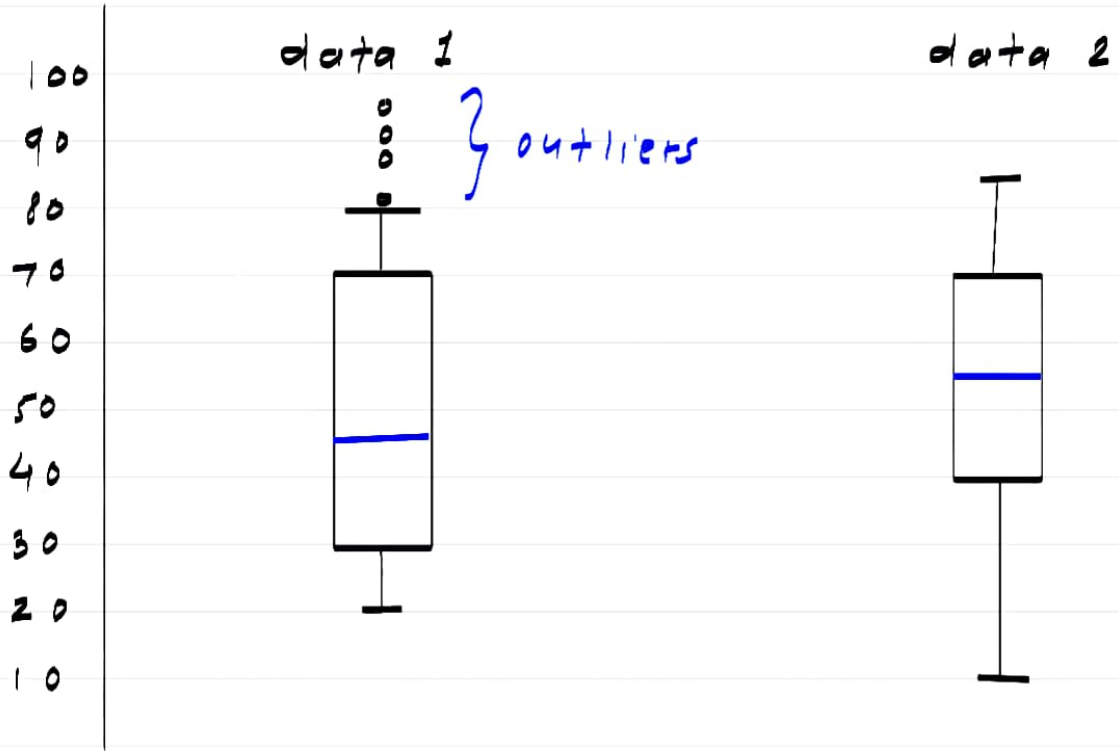
80 matches } A - 45, B - 30, Tied - 5 } data distribution non-visual

$P(A) = \frac{45}{80}$ ✓ $P(B) = \frac{30}{80}$ $P(Tied) = \frac{5}{80}$

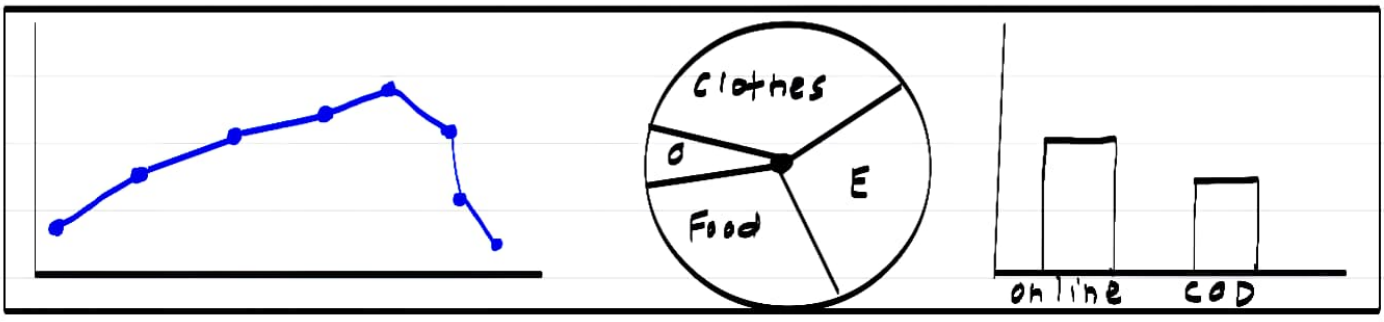
probability - predictive analytics



- Statistical Modelling : Enabling use of correct statistical methods based on the distribution type



using distribution, we can detect outliers and apply correct statistical method to handle them
eg. log-transformation or outlier removal

- Business Applications : User analytics, market trends forcasting outcomes

dashboard - multiple data distributions